

Mahmoud Yahia Ahmed

Generative AI & Conversational AI Engineer | LLMs, RAG & AI Experience Design

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📍 Cairo, Egypt

Summary

Generative AI Engineer with 5+ years of overall experience, including 3+ years focused on Generative AI, LLMs, RAG, NLP, and conversational AI solutions. Experienced in prompt engineering, conversational workflows, AI response optimization, knowledge management, LLM evaluation, guardrails, and AI performance monitoring. Strong ability to translate business requirements into scalable AI solutions and collaborate with business, product, and engineering teams to deliver measurable improvements in automation, accuracy, and user experience.

Profiles

Github	Linkedin	Credly	Portfolio
MahmoudYahia	MahmoudYahia	MahmoudYahia	MahmoudYahia

Education

Ottawa University of Canada [Feb 2022 - Feb 2023] Grade : 9.7 Professional Master of Electrical and Computer Engineering	AlAhram Canadian University [Sep 2016 - July 2020] Grade : 3.27 B.Sc. in Computer Science
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Experience

AIHI Project-based Built a transparency agent for Human Appeal that supports human review, tracks changes to donor contributions, and keeps donors informed through follow-up updates after a short period.	Consultant
Advansys -ForteCloud [Oct 2023 – Aug 2026] Designed and deployed production Generative AI and conversational AI solutions for enterprise use cases, focusing on user journeys, contextual responses, automation, and business requirements.	Senior Machine Learning Engineer Cairo,Egypt
Designed LLM-powered workflows using RAG, prompt engineering, knowledge retrieval, guardrails, and response evaluation to improve accuracy and reduce hallucinations.	
Developed AI applications integrating AWS Bedrock, OpenSearch, Lambda, S3, and DynamoDB, supporting scalable enterprise AI experiences.	
Worked with business and technical stakeholders to translate business requirements into AI workflows, functional requirements, solution designs, and production-ready implementations.	
Implemented evaluation and monitoring approaches to continuously optimize LLM responses, retrieval quality, contextual understanding, and task completion.	
Developed AI-powered document and knowledge assistants enabling users to interact with enterprise information through natural-language queries.	
Developed and optimized prompts through iterative testing and evaluation to improve response quality, contextual understanding, task completion, and hallucination control.	

Route Training Center [July 2023 - Jan 2025] - Designed and led a market-aligned AI curriculum focused on applied ML, NLP, computer vision, and LLM-based systems. - Mentored and supervised 150+ students and junior engineers, enabling delivery of real-world AI projects and improving job-readiness. - Guided capstone projects emphasizing problem formulation, model deployment, and business impact.	Head of Ai & Instructor (part time)
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Microsoft [Sep 2022 - Feb 2023] - Developed a Named Entity Recognition (NER) system for Egyptian Arabic dialects using deep learning. - Built data labeling, augmentation, and training pipelines using Hugging Face Transformers and spaCy. - Improved entity extraction accuracy from noisy transcriptions using phonetic similarity and post-processing heuristics.	Applied NLP
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Up Work [Feb 2021 - Jan 2022] - Delivered over 10 successful ML projects in computer vision, time-series forecasting, and NLP for global clients. - Built custom image classification models, OCR pipelines, and sentiment analysis systems tailored to client business needs. - Collaborated across time zones to deploy models using Flask, FastAPI, and Docker on cloud platforms (AWS, GCP).	Freelancer
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Projects

AI Contract Intelligence Platform Node.js + Terraform + AWS Enterprise Legal Automation Saudi Arabia Government Built a scalable AI-powered contract analysis platform enabling legal and procurement teams to automatically extract obligations, risks, renewal clauses, and compliance insights from enterprise contracts. - Developed backend services using Node.js and Express.js to manage document ingestion, user workflows, and AI orchestration. - Provisioned complete AWS infrastructure using Terraform including ECS services, S3 buckets, IAM roles, Application Load Balancer, CloudWatch monitoring, and OpenSearch vector storage. - Integrated Retrieval-Augmented Generation (RAG) pipelines using Bedrock and OpenSearch for semantic contract querying. - Implemented secure multi-tenant architecture with JWT authentication and role-based access control. - Reduced contract review time by ~65% through automated clause extraction and AI-generated summaries. - Designed scalable infrastructure modules allowing environment replication (dev/staging/prod) in minutes.

Extended the platform with a **self-hosted LLM inference layer on EC2** enabling legal teams to analyze sensitive contracts without exposing documents to third-party model providers; configured environment setup, model serving, and scalable inference endpoints.

Private Enterprise LLM & AI Assistant Platform Designed and deployed a private LLM environment for enterprise AI assistants where data privacy and security restricted the use of external model APIs.
Implemented RAG-based knowledge interaction allowing users to query internal enterprise knowledge through natural-language conversations.
Designed prompt and response strategies, retrieval workflows, and guardrails to improve response quality and reliability.
Optimized model serving, inference latency, and infrastructure cost for production-oriented deployments.

RAG System with LLM Evaluation Pipeline - Built a hybrid search layer combining BM25 sparse retrieval with OpenSearch kNN dense vectors, then added Cohere Rerank API as a cross-encoder re-ranking step to prioritize the most semantically relevant chunks. - Implemented HyDE (Hypothetical Document Embeddings) for query expansion and self-RAG loops where the model evaluates retrieved context quality before answering, falling back to 'I don't know' when confidence is below threshold. - Designed a full evaluation pipeline using RAGAS — tracking faithfulness, answer relevancy, context recall, and context precision — and used GPT-4 as a judge to measure hallucination rate across iterations. - Reduced hallucination rate from 18% '3.2% through citation-grounding prompts, confidence thresholds, and data-driven iteration on chunking strategy and embedding model selection
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Arabic LLM Fine-Tuning — QLoRA & DPO Alignment - Fine-tuned Mistral-7B and Llama-3-8B on a custom 50K-sample Egyptian Arabic dataset using QLoRA (4-bit quantization + LoRA adapters), reducing GPU memory footprint by 60% vs full fine-tuning. - Ran systematic adapter-rank experiments (r = 8, 16, 32) tracking loss, ROUGE, and BERTScore via Weights & Biases to identify the optimal accuracy–compute trade-off. - Applied Direct Preference Optimization (DPO) to align model outputs with human preferences on dialect-specific responses, improving user satisfaction scores by 34% over the base SFT model.

Passport & Government ID Image Validation System UAE Government High-Scale AI System ~50K Requests/Day - Built a computer vision system validate passport and ID images against ICAO standards. - Automated checks for face alignment, white background, shadows, sharpness, pixelation. - Deployed on AWS EC2 using MediaPipe and OpenCV.

Certifications

Azure AI Engineer Associate Microsoft	May 2025	AWS Certified AI Practitioner AWS	July 2025
Azure Data Scientist Associate Microsoft	April 2025	Artificial Intelligence Analyst IBM	April 2022
Big Data IBM	March 2022	Predictive Analytics Modeler IBM	Feb 2022
Business Acumen Dale Carnegie	July 2022		

Skills

Generative AI & Conversational AI

LLMs, Prompt Engineering, Conversational AI, AI Assistants, RAG, Agentic AI, LLM Evaluation, Guardrails, Response Optimization, Knowledge Retrieval, Chatbots

Conversational Experience Design

Conversational Workflows, Dialogue Design, AI Personas, Tone & Response Strategies, Guardrails, Fallback & Escalation Logic, Knowledge Management, Conversation Analytics, AI Performance Optimization

LLM & AI Frameworks

LangChain, LangGraph, AutoGen, Hugging Face, OpenAI

AI/ML Frameworks

NLP, NER, Machine Learning, Deep Learning, Computer Vision, Fine-tuning, LoRA, QLoRA

Cloud & Deployment

AWS, Azure, GCP, Docker, CI/CD, Terraform, EC2, ECS, OpenSearch

Development

Python, SQL, FastAPI, Flask, Java, C/C++

Business & Solution Design

Business Requirements Analysis, AI Solution Design, Process Mapping, User Stories & Functional Requirements, AI Workflow Design, Stakeholder Collaboration

Languages

English

Arabic